Consider the accompanying matrix as the augmented matrix of a linear system. State in words the next two elementary row operations that should be performed in the process of solving the system.

$$\begin{bmatrix} 1 & -6 & 4 & 0 & -2 \\ 0 & 2 & -6 & 0 & 4 \\ 0 & 0 & 1 & 3 & -4 \\ 0 & 0 & 2 & 1 & 12 \end{bmatrix}$$

What should be the first elementary row operation performed? Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

- A. Replace row 4 by its sum with -2 times row 3.
  (Type an integer or a simplified fraction.)
- B. Interchange row 3 and row 2.
- C. Replace row 2 by its sum with ☐ times row 4.
  (Type an integer or a simplified fraction.)
- D. Scale row 1 by ☐.
  (Type an integer or a simplified fraction.)

What should be the second elementary row operation performed? Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

- A. Replace row 3 by its sum with ☐ times row 2.
  (Type an integer or a simplified fraction.)
- B. Replace row 1 by its sum with ☐ times row 4.
  (Type an integer or a simplified fraction.)
- C. Scale row 4 by $-\frac{1}{5}$.
  (Type an integer or a simplified fraction.)
- D. Interchange row 1 and row 2.

Row reduce the matrix to reduced echelon form. Identify the pivot positions in the final matrix and in the original matrix, and list the pivot columns.

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 5 & 6 & 7 \\ 7 & 8 & 9 & 10 \end{bmatrix}$$

Row reduce the matrix to reduced echelon form and identify the pivot positions in the final matrix. The pivot positions are indicated by bold values. Choose the correct answer below.

A. $\begin{bmatrix} \mathbf{1} & 0 & 0 & 0 \\ 0 & \mathbf{1} & 0 & 0 \\ 0 & 0 & \mathbf{1} & \mathbf{1} \end{bmatrix}$

B. $\begin{bmatrix} \mathbf{1} & 0 & 0 & 1 \\ 0 & \mathbf{1} & 0 & 4 \\ 0 & 0 & \mathbf{1} & 7 \end{bmatrix}$

C. $\begin{bmatrix} \mathbf{1} & 0 & -1 & -2 \\ 0 & \mathbf{1} & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

D. $\begin{bmatrix} \mathbf{1} & 2 & 0 & 0 \\ 0 & 0 & \mathbf{1} & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Identify the pivot positions in the original matrix. The pivot positions are indicated by bold values. Choose the correct answer below.

A. $\begin{bmatrix} \mathbf{1} & 2 & 3 & 4 \\ 4 & \mathbf{5} & 6 & 7 \\ 7 & 8 & \mathbf{9} & \mathbf{10} \end{bmatrix}$

B. $\begin{bmatrix} \mathbf{1} & 2 & 3 & 4 \\ 4 & \mathbf{5} & 6 & 7 \\ 7 & 8 & 9 & 10 \end{bmatrix}$

C. $\begin{bmatrix} \mathbf{1} & 2 & 3 & 4 \\ 4 & \mathbf{5} & 6 & 7 \\ 7 & 8 & \mathbf{9} & 10 \end{bmatrix}$

D. $\begin{bmatrix} \mathbf{1} & 2 & 3 & 4 \\ 4 & 5 & \mathbf{6} & 7 \\ 7 & 8 & 9 & 10 \end{bmatrix}$

List the pivot columns. Select all that apply.

- A. Column 2
- B. Column 1
- C. Column 4
- D. Column 3

List five vectors in Span$\{\mathbf{v}_1, \mathbf{v}_2\}$. Do not make a sketch.

$$\mathbf{v}_1 = \begin{bmatrix} 8 \\ 4 \\ -6 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -7 \\ 1 \\ 0 \end{bmatrix}$$

List five vectors in Span$\{\mathbf{v}_1, \mathbf{v}_2\}$.

$$\begin{bmatrix} 8 \\ 4 \\ -6 \end{bmatrix}, \begin{bmatrix} -7 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 5 \\ -6 \end{bmatrix}, \begin{bmatrix} 16 \\ 8 \\ -12 \end{bmatrix}, \begin{bmatrix} 15 \\ 3 \\ -6 \end{bmatrix}$$

(Use the matrix template in the math palette. Use a comma to separate vectors as needed. Type an integer or a simplified fraction for each vector element. Type each answer only once.)

Use the definition of A**x** to write the matrix equation as a vector equation.

$$\begin{bmatrix} 1 & 3 \\ -2 & 6 \\ -1 & -1 \\ -5 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 10 \\ 4 \\ -6 \\ -14 \end{bmatrix}$$

The matrix equation written as a vector equation is $4\begin{bmatrix} 1 \\ -2 \\ -1 \\ -5 \end{bmatrix} + 2\begin{bmatrix} 3 \\ 6 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 10 \\ 4 \\ -6 \\ -14 \end{bmatrix}$.

Describe all solutions of $A\mathbf{x} = \mathbf{0}$ in parametric vector form, where A is row equivalent to the given matrix.

$$\begin{bmatrix} 2 & -4 & 8 \\ -1 & 2 & -4 \end{bmatrix}$$

$$\mathbf{x} = x_2\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + x_3\begin{bmatrix} -4 \\ 0 \\ 1 \end{bmatrix}$$

(Type an integer or fraction for each matrix element.)

Consider an economy with three sectors, Chemicals & Metals, Fuels & Power, and Machinery. Chemicals sells 20% of its output to Fuels and 60% to Machinery and retains the rest. Fuels sells 70% of its output to Chemicals and 20% to Machinery and retains the rest. Machinery sells 40% of its output to Chemicals and 30% to Fuels and retains the rest. Complete parts (a) through (c) below.

a. Construct the exchange table for this economy.

Distribution of Output from:

Chemicals	Fuels	Machinery	Purchased by:
0.20	0.70	0.40	Chemicals
0.20	0.10	0.30	Fuels
0.60	0.20	0.30	Machinery

(Type integers or decimals.)

b. Develop a system of equations that leads to prices at which each sector's income matches its expenses. Then write the augmented matrix that can be row reduced to find these prices. The first, second, and third columns of the matrix should correspond to Chemicals, Fuels, and Machinery, respectively.

The augmented matrix is $\begin{bmatrix} -0.80 & 0.70 & 0.40 & 0 \\ 0.20 & -0.90 & 0.30 & 0 \\ 0.60 & 0.20 & -0.70 & 0 \end{bmatrix}$.

(Type an integer or decimal for each matrix element.)

c. Find a set of equilibrium prices when the price for the Machinery output is 70 units.

$p_{\text{Chemicals}} = 68.8$, $p_{\text{Fuels}} = 38.6$, $p_{\text{Machinery}} = 70.0$

(Type integers or decimals rounded to the nearest tenth as needed.)

If T is defined by $T(\mathbf{x}) = A\mathbf{x}$, find a vector **x** whose image under T is **b**, and determine whether **x** is unique. Let $A = \begin{bmatrix} 1 & -3 & -4 \\ -2 & -1 & -13 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 2 \\ -11 \end{bmatrix}$.

Find a single vector **x** whose image under T is **b**.

$\mathbf{x} = \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}$

Is the vector **x** found in the previous step unique?

- A. No, because there is a free variable in the system of equations.
- B. Yes, because there are no free variables in the system of equations.
- C. No, because there are no free variables in the system of equations.
- D. Yes, because there is a free variable in the system of equations.

Compute the product AB by the definition of the product of matrices, where $A\mathbf{b}_1$ and $A\mathbf{b}_2$ are computed separately, and by the row-column rule for computing AB.

$$A = \begin{bmatrix} -1 & 2 \\ 3 & 5 \\ 5 & -2 \end{bmatrix}, B = \begin{bmatrix} 5 & -2 \\ -3 & 4 \end{bmatrix}$$

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(Type an integer or decimal for each matrix element.)

Set up the product $A\mathbf{b}_2$, where $\mathbf{b}_2$ is the second column of B.

$$A\mathbf{b}_2 = \begin{bmatrix} -1 & 2 \\ 3 & 5 \\ 5 & -2 \end{bmatrix} \begin{bmatrix} -2 \\ 4 \end{bmatrix}$$

(Use one answer box for A and use the other answer box for $\mathbf{b}_2$.)

Calculate $A\mathbf{b}_2$, where $\mathbf{b}_2$ is the second column of B.

$$A\mathbf{b}_2 = \begin{bmatrix} 10 \\ 14 \\ -18 \end{bmatrix}$$

(Type an integer or decimal for each matrix element.)

Determine the numerical expression for the first entry in the first column of AB using the row-column rule. Choose the correct answer below.

- A. $-1(5) + 2(-3)$
- B. $((-1) - (5)) \cdot ((2) - (-3))$
- C. $-1(5) - 2(-3)$
- D. $((-1) + (5)) \cdot ((2) + (-3))$

Determine the product AB.

Let $A = \begin{bmatrix} 3 & 2 \\ -1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 8 \\ -4 & k \end{bmatrix}$. What value(s) of k, if any, will make AB = BA?

Select the correct choice below and, if necessary, fill in the answer box within your choice.

- A. k = -2 (Use a comma to separate answers as needed.)
- B. No value of k will make AB = BA

Find the inverse of the matrix.

$$\begin{bmatrix} 2 & 3 \\ 6 & 5 \end{bmatrix}$$

Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

A. The inverse matrix is $\begin{bmatrix} -\frac{5}{8} & \frac{3}{8} \\ \frac{3}{4} & -\frac{1}{4} \end{bmatrix}$.

(Type an integer or simplified fraction for each matrix element.)

B. The matrix is not invertible.

Find the inverse of the given matrix, if it exists.

$$A = \begin{bmatrix} 1 & -2 & 1 \\ -4 & 9 & -5 \\ 2 & -6 & 4 \end{bmatrix}$$

Find the inverse. Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

A. $A^{-1} =$

(Type an integer or decimal for each matrix element.)

B. The matrix A does not have an inverse.

Solve the equation $A\mathbf{x} = \mathbf{b}$ by using the LU factorization given for A. Also solve $A\mathbf{x} = \mathbf{b}$ by ordinary row reduction.

$$A = \begin{bmatrix} 4 & -5 & -4 \\ -4 & 1 & 3 \\ 8 & 6 & -5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -4 & 1 \end{bmatrix} \begin{bmatrix} 4 & -5 & -4 \\ 0 & -4 & -1 \\ 0 & 0 & -1 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 7 \\ -21 \\ 76 \end{bmatrix}$$

Let $L\mathbf{y} = \mathbf{b}$ and $U\mathbf{x} = \mathbf{y}$. Solve for $\mathbf{x}$ and $\mathbf{y}$.

$$\mathbf{y} = \begin{bmatrix} 7 \\ -14 \\ 6 \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} 2 \\ 5 \\ -6 \end{bmatrix}$$

Row reduce the augmented matrix [A **b**] and use it to find **x**.

The reduced echelon form of [A **b**] is $\begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & -6 \end{bmatrix}$, yielding $\mathbf{x} = \begin{bmatrix} 2 \\ 5 \\ -6 \end{bmatrix}$.

Solve the equation $A\mathbf{x} = \mathbf{b}$ by using the LU factorization given for A.

$$A = \begin{bmatrix} 2 & -2 & -4 \\ 1 & -2 & 7 \\ 3 & -2 & -6 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{3}{2} & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & -2 & -4 \\ 0 & -1 & 9 \\ 0 & 0 & 9 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 0 \\ -5 \\ 14 \end{bmatrix}$$

Let $L\mathbf{y} = \mathbf{b}$. Solve for $\mathbf{y}$.

$$\mathbf{y} = \begin{bmatrix} 0 \\ -5 \\ 9 \end{bmatrix}$$

Let $U\mathbf{x} = \mathbf{y}$. Solve for $\mathbf{x}$.

$$\mathbf{x} = \begin{bmatrix} 16 \\ 14 \\ 1 \end{bmatrix}$$